

ZEN Univ.

LUT 1.



Exercise 2.

$$1. \quad p^3 - kp^2 - 18p = 2k^2 - 3k + 1$$

Factorize both sides.

$$p(p^2 - kp - 18) = (2k - 1)(k - 1)$$

p must divide at least one of

$(2k - 1)$ and $(k - 1)$.

if $p \mid (2^q - 1)$,

$$2^q - 1 = p^k \quad (k \in \mathbb{Z})$$

Quick evaluation, $k \geq 1$ and k is odd.

$$q = \frac{pk+1}{2}$$

Assign this to original equation.

We can finally get the simplified form as:

$$(2-k)p^2 - (1+k^2)p + (k-8b) = 0$$

if $k=1$)

$$p^2 - 2p - 35 = 0$$

$$(p-7)(p+5) = 0$$

so we got $k=1, p=7$

$$n = \frac{pk+1}{2} = 4$$

if $k=3$)

$$p^2 + 10p + 33 = 0$$

and p is positive, so there are
no solutions.

if $k \geq 5$)

$$(k-2)p^2 + (1+k^2)p + (3b-k) = 0$$

(even when $k \geq 39$) $(1+k^2)p$ is faster
increasing than $(3b-k)$, so
there're no solutions.

Now, assume $p \mid (n-1)$.

$$n-1 = pk \quad (2 \leq k \in \mathbb{R}) \quad \text{so} \quad n = pk + 1$$

Here, $k \geq 1$ inferred by $n \geq 1$ and $p \geq 2$

Again, assign this to the equation.

And simplified form we got:

$$(1-k)p^2 - (1+2k^2)p - (1+k) = 0$$

Negate both side,

$$(k-1)p^2 + (1+2k^2)p + (18+k) = 0$$

And we can see left hand side is positive
thus there're no solutions for p .

Finally we got the unique solution

$$\underline{p=9, q=4} \quad \square$$

② Premises : $m, n \in \mathbb{N}$, $m, n > 1$,

$m^2 + n$ and $m + n^2$ are prime.

Let's assume for some prime p ,

$$p \mid \gcd(mn+1, m+n).$$

$$\Rightarrow p \mid (m-1)(n-1).$$

$$(\Leftrightarrow) p \mid (m-1) \quad \vee \quad p \mid (n-1)$$

By symmetry, just assume $p \mid (m-1)$,

$$\text{so } m \equiv 1 \pmod{p}$$

By $mn \equiv 0$, we get $n \equiv -m \equiv -1$

Assign them to $m^2 + n$, $m^2 + n \equiv 1 - 1 \equiv 0$

$m^2 + n$ is prime, so $\underbrace{m^2 + n = p}_{\text{①}}$

By $p \mid (m-1)$ and $m \geq 2$, $\underline{m \geq p}$ ②

① and ② contradicts. ~~we~~

③ Let N be $10^4 - 10^7$.

The problem can be rewritten as

$$\# \left\{ N \cdot \left(\frac{m}{2^k} \right) ; k, m \in \mathbb{Z}, 0 \leq \frac{m}{2^k} \leq 1, 2^k | N \right\}$$

It's enough to check maximum k .

And N can be dividable by 2, 7 times.

So, count m for $0 \leq m \leq 2^9$, the answer is 129.

